Total No. of printed pages = 4

MA 181301 B

17/12/24

2024

B.Tech. 3rd Semester End-Term Examination

MATHEMATICS III-B

Full Marks – 70

Time – Three hours

The figures in the margin indicate full marks for the questions.

Answer question No. 1 and any *four* from the rest.

1. Choose the correct answer : 1×10

   (i) If A and B are two independent random variables these (CO 1)

   (a) $P(A \cup B) = P(A) + P(B)$

   (b) $P(A \cap B) = P(A) \cdot P(B)$

   (c) $P(A \cup B) = P(A) \cdot P(B)$

   (d) $P(A \cap B) = P(A) + P(B)$

   (ii) If $f(x) = cx$, $x = 1, 2, 3, 4$ is a probability mass function then the value of C is (CO 2)

   (a) $\frac{1}{10}$ (b) $\frac{1}{5}$

   (c) $\frac{2}{5}$ (d) $\frac{3}{10}$

   (iii) For any event A if $P(A^c) = \frac{1}{3}$ then $P(A)$ is (CO 1)

   (a) $\frac{2}{3}$ (b) 1

   (c) $\frac{3}{4}$ (d) $\frac{1}{2}$

   (iv) The expectation of sum of two random variables A and B is (CO 2)

   (a) $E(A) + E(B)$ (b) $E(A) \cdot E(B)$

   (c) $E(A)/E(B)$ (d) $E(A) - E(B)$

   where $E(A)$ means expectation of A.

[Turn over

(v) If $A = \begin{bmatrix} a & 0.3 \\ 1/2 & 1/2 \end{bmatrix}$ is PTM then the value of a is (CO 4)

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) 0.6 (d) 0.7

(vi) If $f(x)$ is a probability density function for a continuous random variable X, then the value of $f(c)$ is (CO 3)

(a) 0 (b) 1

(c) $\frac{1}{3}$ (d) $\frac{1}{2}$

where C is a fixed point.

(vii) The median of the values 20, 15, 25, 28, 18, 16, 30 is (CO 5)

(a) 28 (b) 18

(c) 20 (d) 16

(viii) The mode of the series : (CO 5)

6, 5, 3, 4, 3, 7, 8, 5, 9, 5, 4 is

(a) 4 (b) 8

(c) 3 (d) 5

(ix) In a symmetric distribution all moments are (CO 6)

(a) 1 (b) 2

(c) 0 (d) none of these

(x) The co-efficient of skewness usually lies between (CO 6)

(a) -1 and 1 (b) -1 and 0

(c) 0 and 1 (d) none of these

2. (a) An urn contains 10 white and 3 black balls, while another urn contains 3 white and 5 black balls. Two balls are drawn from the first urn and put into the second urn and then a ball is drawn from the later. What is the probability that it is a white ball? 5

(CO 1)

(b) Ten coins are tossed simultaneously. Find the probability of getting at least seven heads. 3

(CO 2)

(c) Find the mean and variance of the Poisson distribution with parameter λ and p having usual notation. 7

(CO 2)

3. (a) The following table gives the number of accidents that took place in an industry during various days in a week. Test whether accidents are uniformly distributed over the week 5 (CO 4)

Day	Mon	Tue	Wed	Thur	Fri	Sat
No.of accidents :	14	18	12	11	15	14

(b) The average marks in Mathematics of a sample space of 100 students was 51 with a standard deviation of 6 marks. Could this have been a random sample from a population with average marks 50? Justify it. 5 (CO 4)

(c) In a normal distribution of marks, 31% are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. 5 (CO 3)

4. (a) Three balls are drawn at random from a box containing 2 white, 3 red and 4 black balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn. Find the joint probability distribution of (X, Y). Also find the marginal distribution of X and Y. 5 + 2 (CO 2)

(b) Prove that for any two independent random variables X and Y, 4 (CO 2)

$$E(XY) = E(X) \cdot E(Y)$$

(c) A problem in statistics is given to three students A, B and C whose chances of solving it are 1/2, 1/3 and 1/4 respectively. What is the probability that the problem will be solved if all of these try independently? 4 (CO 1)

5. (a) The TPM of a Markov claim is $\begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 0 & 1 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix}$ 3 (CO 4)

Draw the transition diagram of three states

(b) Find the first three row moments and central moments of the discrete random variable X with probability mean function : 7 (CO 2)

$$f(x) = \begin{cases} 0.7, & x = 1 \\ 0.2, & x = 2 \\ 0.1, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

(c) The joint distribution of X and Y is given by $P(x, y) = \dfrac{x+y}{21}$; $x = 1, 2, 3$; $y = 1, 2$. Find the marginal distribution of X and Y. Also find $P(X \le 2, Y \le 2)$. 5 (CO 2)

6. (a) Calculate arithmetic mean and median of the distribution : 3 + 2 + 2 (CO 5)

Class interval	Frequency
130–134	5
135–139	15
140-144	28
145–149	24
150–154	17
155–159	10
160–164	1

Also calculate mode.

(b) Calculate the first four moments about mean for the following distribution and hence find β_1 and β_2. 4 + 2 (CO 5)

x :	0	1	2	3	4	5	6	7	8
f :	1	8	28	56	70	56	28	8	1

(c) Give the differences of positive and negative correlation. 2 (CO 5)

7. (a) Give the bivariate data : 4 + 1 (CO 6)

X :	2	4	5	6	8	11
Y :	18	12	10	8	7	5

(i) Obtain the two regression lines

(ii) Calculate Karl Pearson's co-efficient of correlation.

(b) Fit a linear trend of the following by the method of least squares and estimate the production for the year 1986. 5 + 2 (CO 6)

Years :	1978	1979	1980	1981	1982	1983	1984
Production (in quantities) :	40	45	46	42	47	49	46

(c) The first three moments of a distribution about the value 4 are –1.5, 17 and –30. Find the mean and standard deviation of the distribution. 2 + 1 (CO 5)